

Executive Summary

Food additives serve diverse technological functions but differ in safety profiles, evidence quality, and regulatory status. We reviewed nine additive classes (emulsifiers, preservatives, sweeteners, polyols, colors, flavor enhancers, stabilizers, gums, acidity regulators) and key examples within each. All are approved by agencies like EFSA, FDA and JECFA at specified Acceptable Daily Intakes (ADIs) or “ADI not specified” (generally safe) levels, but some remain controversial due to emerging research or vulnerable subpopulations. For each class we summarize typical uses, toxicological data (absorption/metabolism, sensitive groups), evidence strength (animal vs human studies), human outcomes (RCTs/cohorts), dose-response (ADI, NOAEL, threshold), and regulation. We then assign **tiers** (Tier 1: strong concern; Tier 2: moderate; Tier 3: neutral; Tier 4: context-dependent/beneficial) to representative additives with evidence-based rationale. Finally, we present *Bari* logic for automated screening, including algorithmic rules, decision thresholds, required inputs, and example scenarios. A comparative table distills key attributes (function, ADI/NOAEL, tier, main outcomes, regulatory notes) for prominent additives. Our recommendations prioritize primary regulatory sources (EFSA, JECFA, FDA, Codex) and recent meta-analyses; gaps and controversies are highlighted for further research.

Emulsifiers

Function: Emulsifiers (e.g. mono-/diglycerides E471, lecithin E322, polysorbates E432/E436, carboxymethylcellulose E466, carrageenan E407) enable oil–water mixtures (margarine, sauces, ice cream, baked goods) to stay homogeneous [3†L216-L224] [3†L239-L247]. They also improve texture, volume and shelf-life in breads, doughs and confections [3†L216-L224] [3†L237-L245]. Common natural emulsifiers include lecithin (soy/egg) and gums; synthetic ones include polysorbates and polyricinoleates.

Safety (toxicology/ADME): Most emulsifiers are large molecules poorly absorbed. Lecithin hydrolyzes to choline and fatty acids; soy-derived lecithin may carry allergenic protein residues. Polysorbate-80 (P80) yields sorbitol and laurate; carboxymethylcellulose (CMC) is not digested, acting as fiber. ADME: e.g. ~99% of ingested benzoates (relevant for polysorbates’ butyrate link) are metabolized to hippuric acid and excreted [33†L218-L224]; cellulose ethers like CMC pass through gut unabsorbed [23†L75-L82]. No major systemic toxicity has emerged: EFSA and JECFA assign “ADI not specified” or very high ADIs for most (e.g. EFSA group-ADI “not specified” for cellulose emulsifiers [25†]). Vulnerable populations: inflammatory bowel disease (IBD) patients or infants (especially any added to formulas) are sometimes mentioned as more sensitive, but current data are limited.

Evidence strength: Mostly **animal/mechanistic studies**. Mouse studies (Chassaing et al.) show CMC/P80 induce low-grade gut inflammation, altered microbiota, and metabolic syndrome traits [11†L380-L388] [11†L429-L437]. In vitro/ex vivo, CMC and P80 altered human gut microbiota composition and gene expression toward a pro-inflammatory profile [19†L63-L71]. **Human evidence is sparse**. A recent placebo-controlled RCT (60 adults) found that supplementing a low-emulsifier diet with single emulsifiers (CMC, P80, carrageenan, lecithin) *lowered fecal short-chain fatty acids (SCFAs)* but **did not raise** markers of gut inflammation, lipopolysaccharide or metabolic risk [19†L63-L71]. Only carrageenan modestly increased

intestinal permeability. Overall, no direct human pathology from typical emulsifier intake has been demonstrated; evidence is rated **C (limited)** due to few trials.

Human outcomes: In the 2026 RCT, *mean SCFAs* (butyrate, propionate) fell in emulsifier arms, potentially reflecting microbiome shifts 【19†L63-L71】. No significant change in fecal calprotectin, CRP or serum markers vs placebo. A low-emulsifier diet lowered cholesterol (EFD period) 【19†L63-L71】. Observationally, high-processed diets (with many emulsifiers) correlate weakly with IBD or metabolic risk, but confounding factors abound. No adverse effects like allergy or carcinogenesis have been reported at dietary exposure.

Dose considerations: Typical intake is grams/day. EFSA “ADI not specified” implies no foreseeable risk at realistic doses. For P80, regulatory evaluations noted likely limits because 1% in water (as used in mouse studies) is much higher than human food levels. NOspecific human NOAEL; in mice, 1% in water (~7 g/kg) caused inflammation. In human RCT, emulsifier dose via brownies was not explicitly quantified, but likely well below EFSA-approved levels. If we treat ADIs: e.g. none for lecithin (phospholipid), “not specified” for many.

Regulation: EFSA and FDA permit all common emulsifiers under specified uses. EFSA’s Panel has re-evaluated many (e.g. E322 lecithin, E440 pectins) and kept ADIs at “not specified” or set high mg/kg limits 【3†L293-L301】. No bans exist, but any re-evaluation is ongoing (EU systematic re-eval by 2020). Labelling: must list by E-number/name.

Controversies & gaps: Debates center on the gut microbiome. Critics point to animal-model findings and call for caution (e.g. food bloggers note “clean eating” to avoid emulsifiers). EFSA and FDA say available data show no harm in humans; more research is needed on chronic low-dose effects, especially in IBD-prone individuals. The main gap is **lack of long-term human trials**. Also, some acid esters (DATEM, SSL) could release tartaric acid; high free tartarates have laxative potential.

Summary: Emulsifiers improve texture and shelf-life, and are generally considered safe (Tier 3: neutral) under current use. Lecithin (E322) is Tier 4 (benign, adds choline/fatty acids), synthetic surfactants like P80 and CMC are Tier 2 (moderate concern pending more evidence), and carrageenan is borderline Tier 2 due to its disputed GI effects. We assign **Tier 3** to most common emulsifiers, **Tier 2** to those with suggestive inflammatory signals, **Tier 4** to nutritionally beneficial ones (e.g. lecithin), and **Tier 1** to none in this class (no acute toxins).

Preservatives

Function: Preservatives (nitrites/nitrates E249–252, benzoates E210–213, sulfites E220–228, sorbates E200/E202, propionates, etc.) inhibit microbial growth and oxidative spoilage. Nitrates/nitrites preserve meats and give color; benzoates prevent yeast/mold in acidic foods (soft drinks, pickles); sulfites prevent browning and microbial growth (dried fruits, wines, beers). EDTA (E385) sequesters metals to slow oxidation (e.g. in canned seafood).

Safety profile:

- *Nitrites/nitrates:* Nitrate (NO_3^-) is largely excreted but partly converted to nitrite (NO_2^-) via oral bacteria. NO_2^- can oxidize hemoglobin to methemoglobin and form nitrosamines (potential

carcinogens) in the stomach [29†L25-L34] . EFSA's ADI: nitrate 3.7 mg/kg, nitrite 0.07 mg/kg (as nitrite ion) [29†L42-L49] . JECFA same. Typical dietary nitrate is mostly from vegetables; cured meats contribute added nitrate/nitrite. At high exposures, risk is methemoglobinemia in infants ("blue baby"), a known hazard. In adults, conversion to carcinogenic nitrosamines is the main theoretical risk. EFSA concluded that **nitrosamine formation at permitted levels is of low concern**

[30†L13-L22] ; however, it noted uncertainties about long-term effects [30†L13-L22] .

- **Benzoates:** (Benzoic acid/salts) are rapidly absorbed and conjugated with glycine to hippuric acid; >99% excreted [33†L218-L224] . No mutagenicity or teratogenicity at normal doses. EFSA set a group ADI 5 mg/kg (benzoic acid equivalent) [33†L146-L154] (JECFA 0–5 mg/kg). Sensitive individuals may get skin or asthma reactions from benzoates. There is a controversial link to hyperactivity: a UK study showed a **small effect** of certain dye/benzoate mixtures on child behavior [44†L272-L281] , but EFSA deemed the evidence inconclusive and retained ADIs.
- **Sulfites (sulfur dioxide E220 etc.):** Sulfur dioxide and sulfite salts release SO₂, which can cause bronchospasm in asthmatics and destroy thiamine (vitamin B1) in foods. EFSA (2022) found that estimated intakes for high consumers (especially children and adults) might **exceed safe levels**, using margin-of-exposure (MOE) analysis [35†L310-L318] . No ADI was established; instead, EFSA set MOE benchmarks. EFSA noted **animal studies** showing nervous system effects at high doses [35†L319-L324] . Allergic/intolerance reactions (e.g. in asthmatics) are recognized.
- **Sorbic acid (E200) & salts:** Metabolized to CO₂ and water; no absorption issues. Former ADI 0–25 mg/kg (FDA GRAS), updated EFSA ADI 11 mg/kg (as sorbic acid) in 2019 [37†L72-L77] . No major health concerns at normal use.
- **Others:** Propionates (anti-mold in baked goods) have ADI ~10 mg/kg; EDTA is "ADI not specified" by FDA/GRAS.

Strength of evidence: Preservatives have been well-tested in rodents (subchronic/chronic) establishing ADIs. Human evidence mainly comes from observational or case reports. Key RCT/cohort data:

- **Nitrites/nitrates:** Cohort studies link high processed meat intake to cancer risk (colorectal), but confounded by fat, cooking, heme iron. The consensus is that nitrate from veggies is **beneficial** (blood pressure lowering via NO), whereas processed-meat nitrosamines remain debated. EFSA found current ADIs protective [30†L25-L30] .
- **Benzoates:** Lacking RCTs. Ex vivo work: very high benzoate (10×ADI) had **minimal effects** on human gut microbiota composition [33†L253-L262] . Hyperactivity trials (with dyes) showed only slight, inconsistent effects [44†L272-L281] .
- **Sulfites:** Case reports of asthma attacks or anaphylaxis in sensitive individuals. No RCTs.
- **Propionates/Sorbates:** Generally no human concerns reported.

Human outcomes: Aside from niche sensitivities, preservatives at regulated use cause no reported adverse health effects in general populations. High sulfite intake is acknowledged to possibly impair the nervous system [35†L319-L324] , but daily diets rarely approach those levels.

Dose considerations: ADIs govern risk thresholds. For nitrates/nitrites, exceeding ADI (e.g. >20 mg/kg nitrate) over time could allow harmful nitrosamine formation [29†L25-L34] . Benzoates' ADI of 5 mg/kg is not hard to exceed: a 70 kg adult could reach it with ~3.5 g/d, which could occur with multiple soft drinks and pickled snacks. A 2025 study showed that even 35 g/L (10×ADI) benzoate markedly shifts some gut bacteria but not overall diversity [33†L253-L262] .

Regulation: Nitrites and benzoates have specified maximum levels in foods (e.g. ≤ 150 mg/kg for nitrites in meats). Sulfites must be declared if >10 mg/kg and are banned in fresh meat (because of high background nitrate) in some regions. EU labelling requires “contains sulfites” for >10 ppm. US labels sulfite-preserved foods under *Food Allergen* law for asthmatics. Some countries restrict colors and preservatives in baby foods.

Controversies/Gaps: The nitrite–nitrosamine–cancer link is controversial; recent IARC re-evaluations (2021) classify processed meat (with nitrite) as carcinogenic, but largely attribute that to cooking-derived compounds. The beneficial NO from nitrate complicates risk–benefit (at least 800 mg nitrate from vegetables can lower BP). EFSA noted “benefits outweigh risks” of dietary nitrate for cardiovascular health [30†L30-L34]. Sulfites lack data on chronic CNS effects; EFSA called for more study of sulfite neurotoxicity. Benzoate as an antimicrobial may alter gut flora at high dose, but evidence to date suggests minimal impact at realistic intakes [33†L253-L262].

Summary:

- **Nitrate/Nitrite (Tier 2):** Moderate concern due to nitrosamine potential (Tier 2) – EFSA ADIs protective [29†L42-L49] [30†L25-L32].
- **Benzoate (Tier 2):** ADI 5 mg/kg, slight sensitivity/allergy risk, controversial hyperactivity link (Tier 2) [44†L272-L281] [33†L146-L154].
- **Sulfites (Tier 1):** Asthma/thiamine loss and EFSA’s MOE signal for high consumers (Tier 1) [35†L310-L318] [35†L319-L324].
- **Sorbate (Tier 3):** ADI 11 mg/kg, little evidence of harm – neutral (Tier 3).
- **Propionate, EDTA (Tier 3):** No red flags, Tier 3.

Artificial Sweeteners

Function: Artificial (non-nutritive) sweeteners (e.g. aspartame E951, saccharin E954, sucralose E955, acesulfame K E950, neotame E961, advantame) provide high sweetness intensity with minimal calories. Steviol glycosides (E960, from *Stevia rebaudiana*) are plant-based sweeteners. These enable “diet” foods/beverages for calorie reduction or blood sugar control.

Safety profile:

- **Aspartame:** Metabolized to phenylalanine, aspartic acid, and methanol. Persons with phenylketonuria (PKU) cannot metabolize phenylalanine; thus products must warn. JECFA/EFSA ADI: 0–40 mg/kg [39†L238-L241] (e.g. a 70 kg adult could consume ~9–14 diet sodas/day without exceeding ADI [39†L271-L277]). IARC (2023) classified aspartame as Group 2B (possibly carcinogenic) based on “limited” evidence for liver cancer [39†L238-L241] [39†L265-L274]. JECFA reaffirmed ADI; WHO noted “safety not a major concern at common doses” but called for more studies [39†L238-L241] [39†L271-L278].
- **Sucralose:** Mostly excreted unchanged; tiny percent metabolized to chlorinated compounds. ADI 0–15 mg/kg (JECFA). Early concerns about microbiome/gut and glucose tolerance have been raised in rodents, but human RCTs show sucralose does not significantly alter glycemic response in healthy adults.

- *Saccharin*: Excreted unchanged. ADI 0–5 mg/kg. Past rodent bladder-tumor findings at high doses are not relevant to humans (via species-specific mechanisms); deemed safe by IARC and others.
- *Acesulfame K*: ADI 0–15 mg/kg. No specific human issues reported.
- *Stevia (Steviol glycosides)*: ADI 0–4 mg/kg (expressed as steviol); largely non-toxic, some bitter aftertaste at high dose.
- *Others (neotame, advantame)*: Very high sweetness power, very low ADIs, used in limited foods.

No toxicities such as organ damage are known at approved levels. Some claims of migraines or metabolic effects remain unsubstantiated.

Evidence strength: Many RCTs and meta-analyses exist:

- A 2025 meta-analysis of RCTs ($n \approx 1457$) found **no significant differences** in weight, glucose, or lipid outcomes between adults consuming artificially sweetened beverages (ASBs) versus unsweetened controls [42†L349-L358] . This suggests metabolic neutrality in the short term.
- Other meta-analyses show modest weight reduction when replacing sugar with nonnutritive sweeteners, but effects are small. Observational studies sometimes associate ASBs with obesity or diabetes, but are likely confounded by reverse causality (overweight people switch to diet drinks) [42†L349-L358] .
- Aspartame's link to cancer: IARC's classification is based on limited human data (two cohort signals for hepatocellular carcinoma) [39†L265-L274] . No RCT addresses long-term cancer risk. JECFA sees no need to lower ADI [39†L271-L277] .

Human outcomes: Glucose/insulin: In RCTs, artificial sweeteners yield lower postprandial glycemia than sugar and have negligible insulin secretion, potentially beneficial for diabetics. *Gut microbiota*: Some rodent studies show dysbiosis with certain sweeteners, but human data are mixed. A small study found sucralose altered gut microbiome at very high doses; others found no effect. **Body weight:** Replacing sugar with sweeteners may aid weight maintenance in the short term; long-term efficacy is unclear. **Neurological/other:** Aspartame (which yields methanol) has been tested in many RCTs without finding evidence for headache or cognition issues. A few individuals report headaches or allergic-like reactions, but trials often show placebo-level effects. **Safety signal:** Phenylketonurics (0.5% of population) must avoid aspartame. There are no proven adverse effects on pregnancy, children, or other vulnerable groups from acceptable intakes, aside from the PKU issue.

Dose: Typical exposures are well below ADIs. For example, two cans of diet soda (~180 mg aspartame) are <5% of a 40 mg/kg ADI for a 70 kg person [39†L271-L277] . Gestational exposure is not a concern at dietary levels except PKU risk.

Regulation: All major agencies have approved these sweeteners. Aspartame and saccharin are one of the oldest approvals; newer ones (acesulfame, sucralose) went through modern risk assessments. Sweeteners are generally permitted in foods and beverages with concentration limits. Stevia is also authorized (E960) in EU and US. Labels must indicate presence of artificial sweeteners, and in EU products with steviol glycosides must state "with sweetener".

Controversies/Gaps: The biggest controversy is cancer risk of aspartame. The IARC 2023 listing (Group 2B) has raised public concern, though JECFA and regulators maintain the approved ADI is safe [39†L238-L241] [39†L271-L277] . More epidemiological studies and mechanistic research on aspartame's metabolite

methanol (and formaldehyde) are underway. Another gap is long-term metabolic effects: whether lifelong use influences appetite regulation or glucose tolerance. The 2025 meta-analysis suggests no overt harm [42†L349-L358] , but specific sweeteners differ and were pooled. Rare case reports of allergic reactions (aspartame, saccharin, stevia) lack strong evidence.

Summary: Most artificial sweeteners are **Tier 3 (neutral)** with no proven human harm at regulated doses, except:

- **Aspartame (Tier 2):** PKU contraindication and IARC 2B flag.
- **Sucralose (Tier 2):** Emerging scrutiny over microbiome (flagged moderate).
- **Stevia glycosides (Tier 4):** natural origin, may improve glycemic control – borderline beneficial context.
- **Saccharin/acesulfame (Tier 3):** no major concerns.
- Regulatory bodies (EFSA, FDA) uniformly consider them safe within ADIs.

Sugar Alcohols (Polyols)

Function: Polyols (e.g. sorbitol, xylitol, mannitol, erythritol, maltitol) are carbohydrate alcohols used as bulk sweeteners and humectants. They provide sweetness with lower calories (except maltitol) and do not spike blood sugar. Sorbitol and mannitol occur naturally in fruits; others are industrially produced. Common in “sugar-free” candies, gum, diabetic foods.

Safety: Polyols are partially absorbed and metabolized (e.g. sorbitol → fructose in liver); the unabsorbed portion is fermented in the colon. Excess intake leads to osmotic diarrhea, flatulence, and bloating. EU mandates “excess may cause laxative effect” labels. No systemic toxicity. ADIs are generally “not specified” (safe) because toxic effects only appear at high doses. Vulnerable groups: Irritable bowel syndrome (IBS) patients may be sensitive (FODMAP content). Xylitol is extremely toxic to dogs, but harmless to humans.

Evidence: RCTs on polyols focus on GI tolerance: threshold is typically 20–40 g/dose (e.g. one might get diarrhea if >30 g sorbitol). They can cause GI upset in those unaccustomed or with IBS. However, they don't raise blood glucose; some meta-analyses find polyol diets improve dental health (non-cariogenic). No studies show serious long-term harm at moderate use.

Dose: A single package of sugar-free gum (10 pieces) may contain ~1–2 g xylitol; so one would need dozens for GI effects. Maltitol candies often warn about laxative risk if >30 g/day. There is no formal ADI, but JECFA established that daily consumption up to ~10 g is without effect; 20 g causes mild effects, 40 g significant GI upset.

Regulation: All are approved as food additives (e.g. sorbitol E420). Labeling rules require “excessive consumption may cause laxative effects”. No restrictions beyond these warnings.

Controversies: Polyols themselves are benign. The main issue is consumer awareness of the laxative threshold and the potential of hidden polyols in “healthy” foods. Also, polyols can disrupt low-carb (ketogenic) diets if not tracked.

Summary: Polyols are **Tier 3 (neutral)**: safe and useful, though **Tier 2** (caution) for high intakes or IBS-prone individuals. Some (xylitol) have dental benefits (Tier 4 effect). No additive in this class merits Tier 1 concern.

Food Colors

Function: Additives like Tartrazine (E102, Yellow 5), Sunset Yellow (E110, Yellow 6), Allura Red (E129, Red 40), Brilliant Blue (E133), and natural pigments (beta-carotene E160a, annatto E160b, paprika E160c, chlorophyll E140) provide or restore color. They enhance appeal and compensate for loss of color due to processing.

Safety: Synthetic azo dyes have been scrutinized for hypersensitivity (especially urticaria/asthma in salicylate-sensitive people) and neurobehavioral effects. ADIs (EFSA): Tartrazine 7.5, Sunset 5, Allura 7 mg/kg. EFSA found no carcinogenic or genotoxic risk at permitted uses. Ponceau 4R (not in US) had equivocal bladder tumor data in rodents but ADI set at 0.7 mg/kg. Natural colors (carotenes, anthocyanins, paprika) have ADIs “not specified” or high (beta-carotene 5 mg/kg).

Evidence: The Southampton study (McCann 2007) found that mixtures of six artificial colors plus benzoate modestly increased hyperactivity in a general-child cohort [44†L272-L281]. EFSA re-evaluations (2008) acknowledged “limited evidence of a small effect in some children” [44†L272-L281] but overall ADIs were upheld. No clear link to ADHD or cognitive deficits in population studies. Other RCTs (e.g. in preschoolers) showed no consistent effect of single dyes. Except for hypersensitivity, no human morbidity has been linked to dyes.

Dose: People rarely exceed ADI from diet alone, unless very heavy consumers of colored candies/beverages. For example, Tartrazine ADI (7.5 mg/kg) is ~525 mg/d for 70 kg adult. A typical soda contains ~20 mg per can, so 25 cans would be needed to reach ADI.

Regulation: EU requires health warning label (in black print) on products containing E102, E104, E110, E122, E124, E129: “may have an adverse effect on activity and attention in children.” US has no such warning. Some countries ban specific dyes (e.g. Norway banned Yellow 5, Yellow 6; certain red dyes banned in Singapore). Codex Alimentarius permits them with limits (often following EFSA).

Controversies: The hyperactivity link remains debated. Critics push for removing artificial dyes from children's diets as a precaution. EFSA and FDA say current uses are safe for the general population [44†L272-L281]. An unresolved gap is the effect of exposure in sensitive subpopulations, and combined effects of multiple dyes and additives.

Summary: Synthetic colors are **Tier 2 (moderate concern)** due to possible hyperactivity in sensitive children. Benzoate and color mixtures got a small warning in EFSA's evaluation [44†L272-L281]. Allura Red and Tartrazine are Tier 2. Natural colors (e.g. carotenes, anthocyanins) are Tier 4 (beneficial antioxidant context). Overall, color additives are regulated to ensure ADIs are not exceeded; hyperactive-susceptible children are a noted at-risk group (Tier 2 concern).

Flavor Enhancers

Function: Substances like Monosodium Glutamate (MSG, E621) and disodium 5'-guanylate (E626) and 5'-inosinate (E631) enhance umami/savory taste. Yeast extracts (rich in glutamate) serve similarly. They allow reduced use of salt or fat while maintaining flavor.

Safety: MSG is extensively studied. Early anecdotal "Chinese Restaurant Syndrome" (headache, flushing) was not confirmed in controlled trials (only extremely high IV doses caused transient symptoms). JECFA ADI: "not specified" (safe) at normal dietary use. MSG is metabolized as natural glutamate; it is a neurotransmitter precursor but does not cross the blood-brain barrier in normal amounts. Disodium 5'-nucleotides similarly have no known toxicity.

Evidence: No high-quality RCT has linked MSG in foods to chronic disease. Meta-analyses of headaches find no effect at normal exposures. Animal studies using massive doses showed no organ toxicity. Some nutritionists argue MSG may trigger appetite; others find that it can reduce salt use and sodium intake (potential benefit).

Human outcomes: MSG in food causes no clear symptoms in most people. Rare idiosyncratic reactions (flushing, nausea, sweating) can occur, but double-blind studies showed these only at high intakes (~3–5 g, much above culinary levels) and even then inconsistently. It does slightly raise plasma glutamate, but homeostasis clears it quickly. There is no evidence linking MSG to obesity or diabetes in humans; some cohort studies show neutral or even protective associations (due to lowered salt).

Dose: Typical consumption is 0.5–1 g/day in cuisines using it (much lower than eg. dietary protein-bound glutamate). No toxicity threshold found in animal studies up to 10% diet (unrealistic). One concern is the **non-enzymatic reaction** of MSG in prepared foods might produce trace amounts of carcinogenic compounds (e.g. trace 3-deoxyglucosone), but evidence is speculative.

Regulation: MSG is generally recognized as safe (GRAS by FDA) and authorized worldwide. It must be labeled as "monosodium glutamate" (US) or "flavour enhancer E621" (EU). In Sweden, MSG once had warning labeling; now removed. Disodium nucleotides and yeast extracts are uncontroversial authorized additives.

Controversies: A few individuals claim MSG sensitivity; this remains unproven at normal dietary levels. Some believe free glutamate triggers gut growth or endocrine effects, but studies do not support this at food doses. The World Health Organization did not list MSG in any "hazard" category. Research gaps: long-term effects on metabolism or neurodegeneration have not been meaningfully studied in humans.

Summary: MSG and related enhancers are **Tier 3 (neutral)**. No regulatory body views them as toxic. We classify MSG as Tier 3 (neutral) given lack of credible harm, with a note that very high intake anecdotes place it near Tier 2 if someone self-identifies as sensitive (but this is rare).

Stabilizers and Gums

Function: These are thickening or gelling agents that stabilize textures. Stabilizers (often overlap with emulsifiers) include pectin (E440, fruit jams), alginate (E401 sodium alginate, dairy gels), agar (E406,

vegetarian gelling), and carrageenan (E407, dairy analogs). Gums (E412–E425) like guar (E412), xanthan (E415), gum arabic (E414), locust bean gum (E410) add viscosity and bind water. They prevent syneresis in yogurts, thicken soups, or suspend particulates in sauces.

Safety: Most are indigestible polysaccharides similar to dietary fiber. They undergo fermentation in the colon. At high intakes they may cause GI symptoms (e.g. guar gum can cause diarrhea). Pectin and alginate are widely considered fiber supplements (e.g. pectin 20 g/day reduces cholesterol). No toxic metabolites. Regulatory bodies often do not specify ADIs (“ADI not specified” for many).

Carrageenan is controversial: high-molecular-weight carrageenan (food-grade, λ - or κ -carrageenan) is considered safe, but degraded carrageenan (poligeenan, not used in food) caused tumors in rodents. Some animal studies give high doses of food-grade carrageenan and reported gut inflammation and possible promotion of tumors in the presence of other carcinogens. EFSA reviewed carrageenan several times and kept it authorized, though a formal re-evaluation for infants was underway.

Evidence: Human studies: pectin lowers lipids (Cochrane review); no adverse reports. Alginate used for reflux, safe. Xanthan and guar gum slightly lower postprandial glucose and have prebiotic fiber effects in RCTs. Carrageenan: a few small trials of carrageenan-containing yogurts showed no GI harm in healthy people (even some anti-HP infection trials). Meta-analyses: none have linked stabilizers/gums to chronic disease. A 2025 ex vivo trial (see emulsifiers) showed carrageenan raised permeability.

Dose: Food uses are low (0.1–1% of product). For example, 5 g daily of guar is common in weight-loss regimens; at 30 g can cause diarrhea. No formal NOAEL established (they generally have wide safety margins).

Regulation: All common gums/stabilizers are approved (e.g. guar E412, xanthan E415, pectin E440). Maximum use levels are specified (e.g. up to ~15 g/kg in some confectionery). EU mandates labeling (“stabilizer: E440”). In the US they may appear under generic names (e.g. “pectin”, “guar gum”). There is no ADI; safety is assumed at GRAS levels.

Controversies: Carrageenan’s safety is debated in natural-health communities (some advise avoiding it, calling it an “inflammatory” agent). Scientific bodies (EFSA, FDA, JECFA) find no current evidence to restrict it. The need to re-evaluate carrageenan with modern methods is recognized. Some also question gum arabic’s occupational allergy in manufacturing (not food risk).

Summary: Most stabilizers/gums are **Tier 4 (beneficial/context-dependent)** due to their fiber-like properties. Pectin, xanthan, guar, alginate improve texture and may offer health benefits (fiber). Carrageenan is **Tier 2** (moderate) pending human data on chronic inflammation; other gums are Tier 3/4. No Tier 1 agents here.

Acidity Regulators

Function: Acidity regulators (acids and their salts, e.g. citric acid E330, malic E296, lactic E270, fumaric E297, phosphoric acid E338) control pH and add tartness. They preserve foods by lowering pH (inhibiting microbes) and adjust flavor (e.g. citric acid in beverages, phosphoric acid in colas, lactic acid in fermented foods). Carbonates and bicarbonates (E500) are also acidity regulators (raising pH).

Safety: Most are naturally occurring organic acids with low toxicity. Citric, malic, lactic, acetic acids have no ADIs (ADI “not specified”) and are normal metabolites. Phosphoric acid has been scrutinized: excessive dietary phosphate (from cola drinks or processed foods) may affect bone mineral density and kidney health. EFSA set a safe upper intake for total phosphorus (OOP) in 2019, but specifically for additives, uses are generally within expected diets. Excess ingestion of any strong acid can erode dental enamel (e.g. frequent sour candy ingestion), but regulated use is considered safe.

Evidence: There is minimal evidence of harm. Some studies link high cola (phosphoric acid) intake to lower bone density in older women, but these diets also replace milk. Rare case of lactic acidosis from massive ingestion of baking soda (sodium bicarbonate) exists, but irrelevant to normal use. A small proportion of people may experience GI upset (heartburn) if consuming large acidic drinks.

Dose: These acids are generally consumed in grams: e.g. a lemonade might have 1–2% citric acid (~1–2 g per 250 ml). ADI “not specified” suggests no risk at any realistic consumption. Phosphoric acid in soda is ~50 mg/100 ml, so 500 ml cola provides ~250 mg phosphorus, well below the daily RDI (~700 mg P in diet).

Regulation: All are universally permitted, often without numerical limits (using “quantum satis” or broad categories). Labeling may indicate “acidity regulator (citric acid)”. No restrictions beyond normal labeling; these are seen as GRAS/ESAs.

Controversies: The main discussion is around soft drinks: phosphoric acid consumption vs bone health and CKD (chronic kidney disease). Evidence is mixed, and confounders exist (caffeine, lifestyle). EFSA and FDA do not view these acid levels as risky. Non-acidic alternatives (citric acid) have replaced some phosphoric use due to health perceptions. People with GERD might avoid citrus acids.

Summary: Acidity regulators are **Tier 3/4**: essentially safe and even beneficial for flavor and preservation. Citric, malic, lactic, acetic, bicarbonates are Tier 4 (natural/benign). Phosphoric acid is Tier 3 (neutral) unless consumed extremely (like daily liters of cola). There are no Tier 1 concerns in this class.

Proposed Tier Classification

Based on the above evidence, we propose:

- **Tier 1 (Strong Concern):** *Sulfites (E220–228)* – due to EFSA’s finding that high consumers (esp. children) may approach unsafe exposure, and risk of asthmatic reactions **【35†L310-L318】** . No other group merits Tier 1 in current data.
- **Tier 2 (Moderate):**
 - **Nitrates/Nitrites (E249–252)** – theoretical nitrosamine cancer risk, balanced by cardio benefits; ADIs but controversial scientific debate **【29†L42-L49】 【30†L13-L22】** .
 - **Benzoates (E210–213)** – ADI 5 mg/kg; mild allergen/hypersensitivity, and a loose link to hyperactivity (EFSA noted small effect with dyes **【44†L272-L281】**).
 - **Aspartame (E951)** – PKU risk plus IARC 2B cancer signal **【39†L238-L241】** .
 - **Polysorbate-80, CMC (E432, E466)** – emulsifiers with animal-inflammation data, pending human clarity (Tier 2).

- **Carrageenan (E407)** – potential GI inflammation from animal data; flagged moderate pending new data.
- **Artificial Colors (E102, E110, E122, E124, E129)** – possible ADHD effect in sensitive kids (as warned by EFSA) [44†L272-L281] .
- **Tartrazine, Sunset Yellow, Allura Red** specifically would be Tier 2, as example dyes.
- **Tier 3 (Neutral):**
 - **MSG (E621)** and other glutamates – widely deemed safe (Tier 3).
 - **Sorbates (E200-202), Propionates** – ADIs exist, no significant concerns; Tier 3.
 - **Xylitol, Erythritol, Sorbitol (Polyols)** – harmless at moderate intake; Tier 3 (caution only for GI).
 - **Saccharin, Sucralose, Acesulfame (E950, E955, E954)** – ADIs, minor concerns; Tier 3 (or 2 for sucralose? but given meta-analysis, likely Tier 3).
 - **Pectin, Xanthan (E440, E415)** – fiber-like, safe; Tier 3.
 - **Phosphoric acid (E338)** – possibly Tier 3; moderate evidence of risk only at very high cola consumption.
 - **Gums (guar E412, arabic E414)** – generally benign (Tier 3 or 4).
 - **Natural colors** (beta-carotene E160a, etc.) – Tier 4 (contextual benefit as nutrients).
- **Tier 4 (Beneficial/Contextual):**
 - **Citric acid (E330)** and other natural acids – beneficial for flavor/antioxidant (Tier 4).
 - **Lecithin (E322)** – provides choline; Tier 4.
 - **Bicarbonates (E500)** – neutral/beneficial (leavening); Tier 4.
 - **Xanthan/Guar/Tara (E412, E412)** as fiber – Tier 4 in moderate uses.
 - **Steviol glycosides (E960)** – natural, possibly positive (Tier 4).

This classification is nuanced: within a class, specific additives vary. The summary table below compares key examples across function, ADI, tier and evidence.

Additive (E#)	Class	Main Function	ADI/NOAEL
Tier (Risk Level)	Key Human Evidence	Regulatory	Notes
-----	-----	-----	-----
Emulsifiers:			
Polysorbate 80 (E432)	Emulsifier	Stabilize emulsions	*ns*
(EU); 25 mg/kg†	Tier 2	Animal gut-inflammation (C grade); RCT saw ↓ SCFA [19†L63-L71] . EFSA: safe.	
CMC (E466)	Emulsifier	Thickener/emulsifier	
ns	Tier 2	Animal studies of intestinal inflammation; no human harm known. ADME: not absorbed [23†L75-L82] .	
Lecithin (E322)	Emulsifier	Emulsion stabilizer	
ns	Tier 4	Provides choline/fat; EFSA: no safety	

concern [3†L299-L303] . Generally benign. |
| ****Preservatives:**** | |
| |
| Sodium Nitrite (E250) | Preservative | Meat curing (anti-C. bot) |
0.07 mg/kg (as NO₂⁻) | Tier 2 | EFSA: current ADI protective [29†L42-L49] ; nitrosamine risk low concern [30†L13-L22] . |
| Sodium Nitrate (E251) | Preservative | Meat curing/cheese bloating |
3.7 mg/kg (NO₃⁻) | Tier 2 | EFSA: ADI protective; beneficial
vasodilator at normal levels [30†L30-L34] . |
| Sodium Benzoate (E211) | Preservative | Yeast/mold inhibitor | 0-5 mg/kg
(benzoic) | Tier 2 | ADI 5 mg/kg [33†L146-L154] ; sensitive allergic
individuals exist; minimal gut changes even at high dose [33†L253-L262] . |
| Sulfur Dioxide (E220) | Preservative | Browning inhibitor, antioxidant |
MOE (no ADI) | Tier 1 | EFSA: MOE <80 for high consumers - possible
neurotoxic risk [35†L310-L318] ; asthma trigger. |
| Potassium Sorbate (E202) | Preservative | Mold inhibitor (baked goods) |
11 mg/kg (sorbic) | Tier 3 | EFSA raised ADI to 11 mg/kg (sorbic acid
eq) [37†L72-L77] ; no specific human hazard. |
| ****Artificial Sweeteners:**** | |
| |
| |
| Aspartame (E951) | Sweetener | High-intensity sweetener | 0-40 mg/
kg | Tier 2 | JECFA/EFSA: ADI 40; IARC 2B carcinogen
(limited evidence) [39†L238-L241] ; PKU risk. |
| Sucralose (E955) | Sweetener | High-intensity sweetener | 0-15 mg/
kg | Tier 3 | Safe at ADI; RCTs show no effect on metabolism
[42†L349-L358] ; minimal absorption. |
| Steviol Glycosides (E960) | Sweetener | Natural sweetener | 0-4 mg/kg
(steviol) | Tier 4 | EFSA: ADI 4 mg/kg; no toxicity; possible benefits
(nil glycemic). |
| Saccharin (E954) | Sweetener | High-intensity sweetener | 0-5 mg/
kg | Tier 3 | Long history; rodent tumors irrelevant to
humans; safe at usual doses. |
| ****Polyols:**** | |
| |
| |
| Sorbitol (E420) | Polyol sweetener | Bulk sweetener |
ns | Tier 3 | "Laxative above X g"; no systemic harm
[33†L218-L224] ; diabetic-friendly. |
| Xylitol (E967) | Polyol sweetener | Bulk sweetener |
ns | Tier 3 | Dental benefits; safe for humans (dog
toxin irrelevant); high dose → GI upset. |
| Erythritol (E968) | Polyol sweetener | Bulk sweetener |
ns | Tier 4 | Well-tolerated; excreted unchanged,
minimal GI issues [41†] ; some studies show vascular benefits. |
| ****Food Colors:**** | |

Tartrazine (E102)	Color (yellow)	Colorant	0-7.5 mg/kg
Tier 2	EFSA: ADI 7.5; possible hyperactivity trigger in susceptible kids [44†L272-L281] . Hypersensitivity (rare).		
Sunset Yellow (E110)	Color (yellow)	Colorant	0-5 mg/kg
Tier 2	EFSA: ADI 5; included in ADHD warning label (EU). Possible allergy (less than tartrazine).		
Allura Red (E129)	Color (red)	Colorant	0-7 mg/kg
Tier 2	EFSA re-evaluation in progress; possible ADHD concerns; banned in some countries (e.g. US banned Red 2).		
Beta-Carotene (E160a)	Color (orange)	Colorant (antioxidant)	*ns*
(provitamin A)	Tier 4	Nutrient value; safe as food color.	
Flavor Enhancers:			
MSG (E621)	Umami enhancer	Flavor enhancer (umami)	
ns	Tier 3	GRAS; RCTs show no consistent "sensitivity"; safe for general use.	
Disodium Inosinate (E631)	Umami enhancer	Flavor enhancer	
ns	Tier 3	Used with MSG; no specific toxicity data, considered safe.	
Stabilizers/Gums:			
Carrageenan (E407)	Stabilizer/Gum	Thickener (dairy, plant)	
ns	Tier 2	Animal-inflammation evidence; EFSA keeps it authorized but reviewing.	
Pectin (E440)	Stabilizer/Gum	Gelling agent (jams)	
ns	Tier 4	Soluble fiber; beneficial (lowers cholesterol); no safety issues [3†L293-L301] .	
Xanthan Gum (E415)	Stabilizer/Gum	Thickener (dressings)	
ns	Tier 3	Fermentable fiber; RCTs show minor GI effects at high dose; safe.	
Guar Gum (E412)	Stabilizer/Gum	Thickener (dairy, sauces)	
ns	Tier 3	Hypoallergenic; used as fiber supplement; GI effects at high dose.	
Acidity Regulators:			
Citric Acid (E330)	Acidulant	pH adjustment, flavor	
ns	Tier 4	Natural in citrus; safe; can enhance iron absorption.	
Phosphoric Acid (E338)	Acidulant	pH adjustment (cola)	
ns	Tier 3	Safe at food levels; some studies link high intake to bone changes.	
Sodium Bicarbonate (E500)	Buffer	Acidity regulator / leavening	

ns	Tier 4	Common kitchen chemical; well tolerated in food uses.
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Notes: “ADI/NOAEL” column shows current regulatory values (if *ns*, EFSA lists as “ADI not specified”). Tier reflects our evidence-based concern level. Human outcomes summarize broad findings from RCTs/cohorts and regulatory statements.

Bari Logic and Decision Framework

To implement additive risk assessment in Bari, we propose the following algorithmic structure:

1. Inputs:

2. Additive identity (e.g. “E250 sodium nitrite” or common name)
3. Dose per serving (mg) and consumption frequency
4. User parameters (body weight, special populations like children, pregnancy)
5. **Retrieve Data:** Look up the additive’s class, ADI (mg/kg), tier assignment, key effects, and regulatory status from our database.

6. Decision Rules (pseudocode):

```
FOR each additive in product:
  Tier = lookupTier(additive)
  ADI = lookupADI(additive)
  dailyExposure = dosePerServing * servingsPerDay / bodyWeight
  IF Tier == 1 THEN
    mark HighConcern
  ELSE IF Tier == 2 THEN
    IF ADI exists AND dailyExposure > 0.5*ADI THEN
      mark ModerateConcern
    ELSE IF user has vulnerability (e.g. asthma for sulfites) THEN
      mark ModerateConcern
    ELSE
      mark LowConcern
  ELSE IF Tier == 3 THEN
    mark LowConcern
  ELSE IF Tier == 4 THEN
    mark LowConcern (potentially beneficial)
```

- *Threshold example:* For preservatives like nitrite, use 0.5×ADI as a conservative trigger. For sulfites, any detectable amount for an asthmatic is a red flag.

1. Flowchart (Mermaid):

```
graph TD
    A[Input: additive, dose, frequency, user profile] --> B{Look up Tier}
    B -->|Tier 1| C[Flag: Strong Concern (High)]
    B -->|Tier 2| D{Check Dose \n and Population}
    D -->|Exposure > 0.5*ADI OR vulnerable pop| E[Flag: Moderate Concern]
    D -->|Else| F[Flag: Low/Acceptable]
    B -->|Tier 3| F
    B -->|Tier 4| G[Flag: Beneficial/Contextual (Low)]
    C --> H[Action: Recommend Avoid/Limit]
    E --> I[Action: Caution, possibly limit]
    F --> J[Action: OK within limits]
    G --> K[Action: Positive Note]
```

1. Example Use Cases:

2. *User consumes a soda with 10 mg sodium benzoate per serving, drinks 3 servings/day, weight 70 kg.*

Lookup: Benzoate (Tier 2, ADI 5 mg/kg). Daily dose = 30 mg. $30/(70 \times 5) = 8.6\%$ of ADI. <50%. No vulnerability flagged. **Result:** Tier 2 but exposure low → Low/Moderate concern*.

3. *User with asthma has snack with sulfites (5 mg/serving) 2/day.*

Lookup: Sulfite (Tier 1, no safe ADI). **Result:** Tier 1 → High concern, advise caution or elimination.

4. *Pregnant woman eats yogurt with 200 mg carrageenan (E407).*

Tier: Carrageenan Tier 2 (potential inflammation). 200 mg is low (ADI NS). **Result:** Moderate concern; if GI symptoms, reduce; else ok.

5. *Child weight 20 kg eats candy with Tartrazine 3 mg and benzoate 50 mg.*

Tartrazine: Tier 2, ADI 7.5 mg/kg = 150 mg for child; consumed 3 mg (2% ADI). Benzoate: Tier 2, ADI 100 mg; consumed 50 mg (50% ADI). Exposure flagged for benzoate (exactly 50%). Mixed exposures might be cumulative but handled additively in future versions. **Result:** Tartrazine low concern; Benzoate at threshold → Moderate caution.

6. *User subs for sugar: drinks 2 diet sodas (aspartame 180 mg total), weight 70 kg.*

Aspartame ADI 2800 mg (70×40). Intake 180 mg < 10% ADI. Tier 2 (due to IARC), but exposure trivial. **Result:** Low concern with note on PKU (irrelevant to non-PKU).

Conclusions and Next Steps

Our analysis integrates regulatory benchmarks and current science into a tiered framework. **Tier 1** additives (sulfites) should trigger the strictest scrutiny; **Tier 2** additives (e.g. nitrates, artificial colors, some emulsifiers) warrant moderate caution, especially at high doses or in susceptible people. Most others are Tier 3 or 4 (no action needed beyond normal labels).

For Bari's implementation, we recommend incorporating:

- A database of additives with class, ADIs, vulnerability flags and tier designation.
- Parsing of labels to identify E-numbers/names and quantifying per-serving dose.
- The decision logic as outlined, with user-specific flags (age, weight, sensitivities).
- Regular updates as new evidence (e.g. EFSA re-evaluations, cohort studies) emerges.

Summary Table: The above table compares representative additives on key parameters. It can be expanded or refined, but highlights that **official sources (EFSA, JECFA, WHO) generally consider current uses safe (ADI protective) [29†L42-L49] [39†L271-L278]** . Where evidence is limited or suggestive (e.g. emulsifiers, aspartame, sulfites, dyes), we assign a higher tier and note research gaps.

Implementation Note: All criteria (functions, ADI, tier) should be stored in Bari's database. The pseudocode and flowchart above guide how to use additive identity and dose to produce a risk flag. In practice, Bari could output warnings like "Contains Tier 2 additive(s) at moderate exposure: limit intake" or "Tier 3 additive: safe use" along with rationale. Example: "Beetroot color (Tier 4, beneficial) – no concern" or "Tartrazine (Tier 2) – caution for children (ADI margin ok but potential hyperactivity)".

This framework combines authoritative standards with the latest research findings **[19†L63-L71] [39†L238-L241] [42†L349-L358]** . It is designed to be updated; for any additive not found or newly evaluated, default to Tier 3 and standard ADI checks.
